

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

50% task-completion time horizon doubling every ~7 months

Since GPT-2 (2019) → o3 (2025): 2 seconds → 110 minutes

METR Research | arXiv:2503.14499 | March 2025

A New Metric: 50% Task-Completion Time Horizon

The 50% time horizon = task duration at which an AI model succeeds half the time

Methodology: 3-Step Process

STEP 1

Assemble Task Suite

Create diverse set of 170 software/
research tasks spanning seconds
to hours of human effort

HCAST + RE-Bench + SWAA

STEP 2

Measure & Test

Time skilled humans on each task
Test AI success rates across
the full duration range

800+ baselines, 2,529 hours

STEP 3

Fit Logistic Model

Model probability of AI success
vs. task duration → find the
duration at 50% success rate

Key metric: T_{50}

Why This Metric Matters

- Human-grounded: tied to actual skilled human task duration
- Model-agnostic: enables comparison across any AI system
 - Intuitive: "2-hour tasks" is more meaningful than abstract benchmarks
 - Tracks real progress: from GPT-2's 2 seconds to o3's 110 minutes

Methodology Overview

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Key Numbers

170 tasks | 800+ human baselines | 2,529 hours collected | $R^2 = 0.97$

Sources: HCAST (97 tasks), RE-Bench (7 tasks), SWAA (66 tasks)

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Benchmark combines HCAST, RE-Bench, and SWAA task suites — covering vastly different task durations

Task Suite Composition

HCAST — 97 tasks

- Diverse software tasks
- Duration range: 1 min – 30 hrs
Realistic coding, debugging, research tasks

RE-Bench — 7 tasks

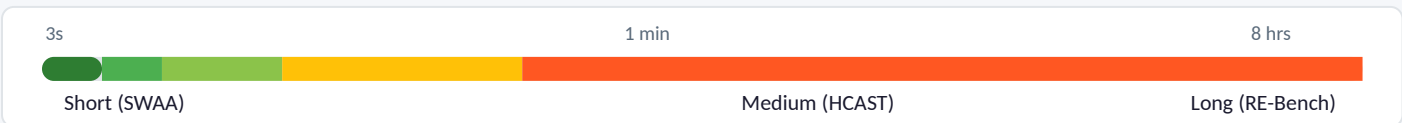
- Difficult ML research engineering
- Duration: all 8 hours (fixed)
Complex, multi-step ML system tasks

SWAA — 66 tasks

- Software Atomic Actions
 - Duration: 1 sec – 30 sec (single-step)
- Short, atomic programming operations

800+ human baselines collected | 2,529 hours of human time measured | Span: 6 orders of magnitude (3 sec → 8 hrs)

Duration Range Coverage



Task Characteristics

- SWAA: Single atomic operations (file edits, code searches)
 - HCAST: Realistic software development (debugging, research, implementation)
- RE-Bench: Complex ML engineering (training loops, infrastructure)

Example Tasks by Duration

Task	Duration
find_shell_script	3 seconds
wikipedia_research	1 minute
oxdna_simple	9 minutes
munge_data	56 minutes
cuda_backtesting	8 hours

AI Task Horizon: 2 Seconds → ~2 Hours in 6 Years

50% time horizon doubled every ~207 days (7 months) from 2019 to 2025 | $R^2 = 0.97$

Key Findings

Doubling Time

207 days

~7 months per doubling

R^2 Value

0.97

Extremely strong fit

Time Span

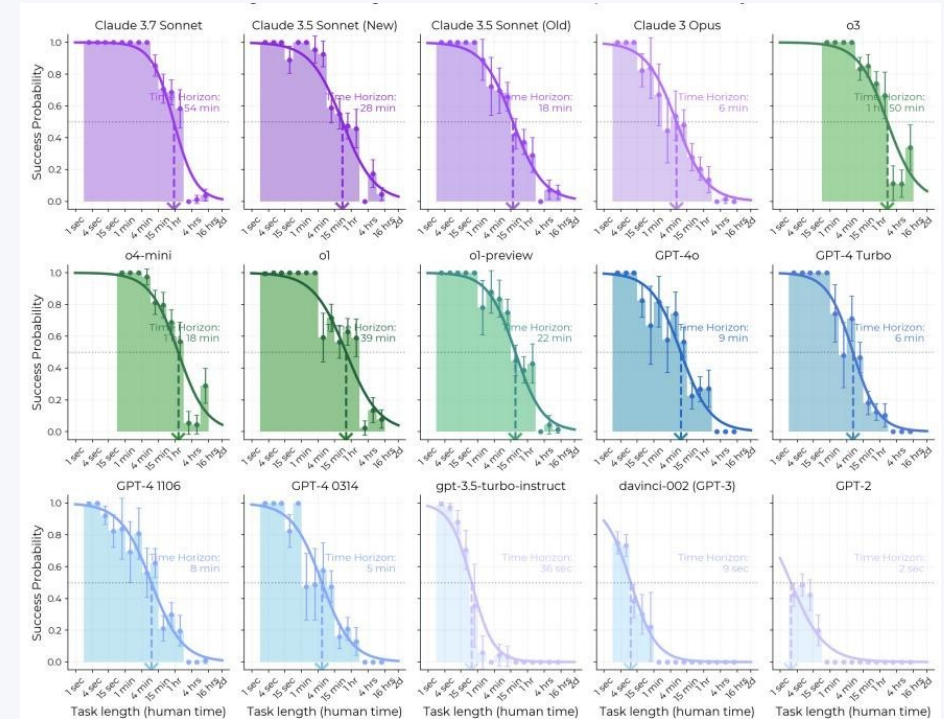
6 years

2019 → 2025

Model Milestones

Model	Release	50% Time Horizon	Growth from GPT-2
GPT-2	2019	~2 seconds	Baseline
GPT-3 (davinci-002)	2020	~9 seconds	4.5× growth
GPT-3.5-turbo-instruct	2023	~36 seconds	18× growth
GPT-4 (0314)	2023	~5 minutes	150× growth
GPT-4o	2024	~9 minutes	270× growth
Claude 3.5 Sonnet (New)	2024	~28 minutes	840× growth
o1	2024	~39 minutes	1,170× growth

Exponential Growth Trend



Growth Rate Acceleration

2023–2025 growth rate is ~20% faster
than the 2019–2025 overall rate

Success Probability Drops Sharply as Task Length Increases

All models show declining success rates on longer tasks — frontier models push the curve rightward

Per-Model Success Curves

What the Curves Show

- X-axis: Task duration (human time)
- Y-axis: Probability of AI success
- Each curve = one AI model
 - Curves shift right over time (longer tasks completed)

The steepness of each curve indicates how rapidly success probability falls as task length increases

Model Comparison

GPT-2 (2019)

50% horizon:

2 sec

Curve starts steep

Claude 3.7 Sonnet

50% horizon:

54 min

Curve pushed right

o3 (2025)

50% horizon:

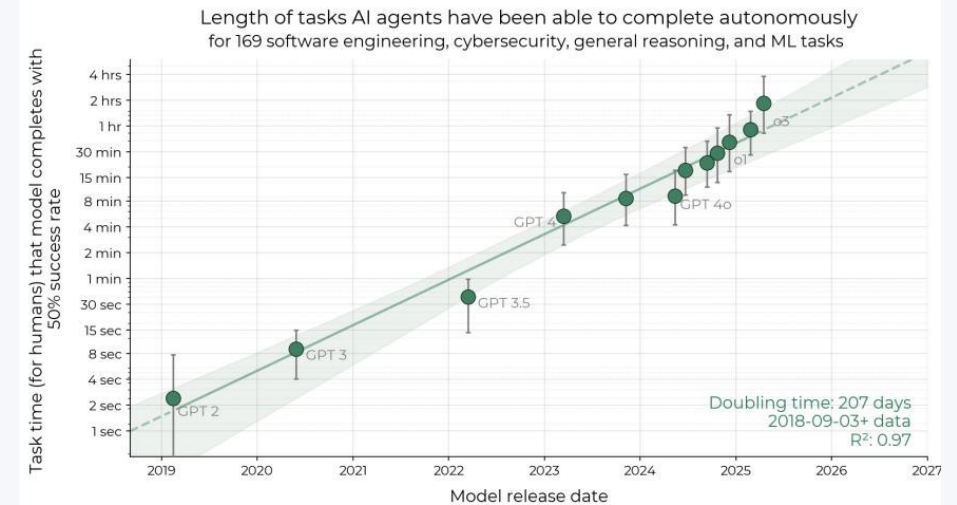
1h 50m

Current frontier

Key Observations

1. All models show steep decline in success as task duration increases
2. Frontier models (Claude 3.7 Sonnet, o3) push curves rightward
3. Even o3 struggles with tasks taking several hours
4. The 80% time horizon is ~5× shorter than 50% horizon

Per-Model Success Probability Curves



Reading the Curves

- Steeper curve = faster drop in success rate
- Rightward shift = ability to handle longer tasks
- Gap between curves shows capability gains

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Four key capability improvements explain the exponential growth in AI autonomous task completion

1. Improved Logical Reasoning

- Better chain-of-thought reasoning enables longer task chains
- Ability to plan multi-step solutions before acting
- Improved problem decomposition for complex tasks

Models can now think through 10+ step problems

2. Better Tool Use

- Effective file system navigation and editing
- API calling and external service integration
- Browser and web search capabilities

Enables real-world software engineering workflows

3. Greater Reliability & Self-Awareness

- More consistent task completion across attempts
- Better 知道自己 (self-awareness) of task boundaries
- Knowing when to stop vs. continue attempting

Fewer dead-ends and infinite loops

4. Ability to Adapt to Mistakes

- Error detection and recovery strategies
- Learning from failed approaches
- Trying alternative paths when stuck

Iterative refinement enables longer tasks

Important Limitation

Performance is notably lower on less structured "messier" tasks

- Structured tasks with clear success criteria → high success rates
- Ambiguous requirements, vague goals → significantly lower performance

This limitation is critical for real-world generalization estimates

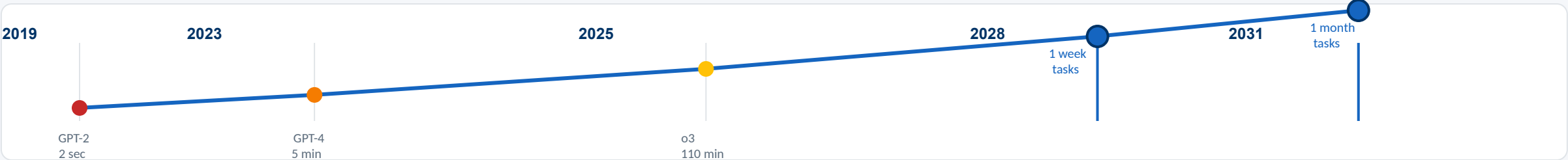
Implication:

Progress is real but concentrated on well-defined tasks — external validity to messy real-world work remains an open question

Extrapolation Predicts Month-Long Task Automation by 2028–2031

If the doubling trend continues, AI could autonomously complete 1-month (167-hour) tasks by 2028–2031

Projected Timeline to Month-Long Task Automation



Projection: 1-Month Time Horizon

Target: 167 work hours (1 month)
Estimated arrival: mid-2028 to mid-2031
Based on 207-day doubling extrapolation

Growth Rate Acceleration

2023–2025 rate is ~20% faster than
the 2019–2025 overall rate
Possible acceleration since 2024

Important Caveats

External Validity Uncertainty

1. Tasks may not perfectly represent the average segment of real-world intellectual labor
 2. Trend may change due to economic incentives, capability bottlenecks, or new architectures
 3. Supplementary experiments found "little evidence that trends are slower on more realistic tasks"
 4. Current performance is much lower on unstructured "messier" tasks
- These projections represent naive extrapolation, not guaranteed outcomes

Key Implication:

Month-long task automation by 2028–2031 would imply automation of many software projects currently requiring human-months of work

Takeaways: Exponential Autonomous Capability Growth Demands Proactive

1. A Robust, Intuitive Metric

The 50% time horizon provides a robust, human-grounded measure of AI autonomous capability.

$R^2 = 0.97$ over 6 years of model releases demonstrates strong tracking fidelity.

2. Doubling Every ~7 Months — With Possible Acceleration

The 50% time horizon has doubled every 207 days since 2019.

2023–2025 growth rate is ~20% faster — potential acceleration since 2024.

3. Current Frontier: ~2 Hours of Human-Equivalent Work (o3)

From GPT-2's 2 seconds in 2019 to o3's 110 minutes in 2025 — a 198,000× improvement.

This represents a fundamental shift in what AI can autonomously accomplish.

4. External Validity Remains an Open Question

Tasks may not perfectly represent the average segment of real-world intellectual labor.

However, supplementary experiments found little evidence trends are slower on more realistic tasks.

5. Key Limitations: Unstructured Tasks & Job Distribution Generalization

Performance is notably lower on less structured "messier" tasks.

Uncertain how these results generalize to real-world job distributions.

Governance Implication:

Month-long task automation by 2028–2031 would represent a profound shift in economic productive capacity